A CANONICAL ISOMORPHISM BETWEEN THE DETERMINANT OF THE MIDDLE TERM OF A SHORT EXACT SEQUENCE OF FINITE PROJECTIVE MODULES AND THE TENSOR PRODUCT OF THE OUTER DETERMINANTS

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Editorial note. The proof environment is opened by the author as \begin{proof}[First proof] and the author writes 'We omit the detailed proof that this is well defined and an isomorphism'; the environment and that sentence are kept verbatim.

Allowed standard prerequisites (editorial).

- Determinant (top exterior power) of a finite projective module and local constancy of rank. the notation det in the statement at source lines 36291-36292 and the rank decomposition at proof lines 36296-36299. Supplies the two sides of the required isomorphism and the product decomposition that makes the ranks constant on each factor.
- Exterior powers of modules and induced maps on top exterior powers. the map wedge^i(M') tensor wedge^j(M") -> wedge^{i+j}(M) at proof lines 36301-36305. Is the explicit form the canonical isomorphism takes once the ranks are constant.

Lemma 0.2. *Let R be a ring. Let*

$$0 \to M' \to M \to M'' \to 0$$

be a short exact sequence of finite projective R-modules. Then there is a canonical isomorphism

$$\gamma : \det(M') \otimes \det(M'') \longrightarrow \det(M)$$

First proof. First proof. Decompose R into a product of rings R_{ij} such that $M' = \prod M'_{ij}$ and $M'' = \prod M''_{ij}$ where M'_{ij} has rank i and M''_{ij} has rank j. Of course then $M = \prod M_{ij}$ and M_{ij} has rank $i + j$. This reduces us to the case where M' and M'' have constant rank say i and j. In this case we have to construct a canonical map

$$\wedge^i(M') \otimes \wedge^j(M'') \longrightarrow \wedge^{i+j}(M)$$

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To do this choose $m'_1, \ldots, m'_i$ in M' and $m''_1, \ldots, m''_j$ in M''. Denote $m_1, \ldots, m_i \in M$ the images of $m'_1, \ldots, m'_i$ and denote $m_{i+1}, \ldots, m_{i+j} \in M$ elements mapping to $m''_1, \ldots, m''_j$ in M''. Our rule will be that

$$m'_1 \wedge \ldots \wedge m'_i \otimes m''_1 \wedge \ldots \wedge m''_j \longmapsto m_1 \wedge \ldots \wedge m_{i+j}$$

We omit the detailed proof that this is well defined and an isomorphism. $\square$